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Traffic Demand Forecasting: Stacked Ensemble Methodology

This project utilizes a stacked ensemble architecture to predict traffic demand. By combining multiple gradient-boosting models with a Bayesian meta-learner, the pipeline minimizes prediction error, reduces variance, and ensures robust generalization to unseen data.

Why this approach?

We adopted a stacking ensemble strategy to move beyond the limitations of single-model architectures. The rationale for this design includes:

- **Robustness via Model Diversity:** CatBoost and XGBoost utilize fundamentally different tree-construction algorithms (oblivious vs. leaf-wise). By combining them, we create a system where the strengths of one model mitigate the specific biases of the other.
- **Prevention of Overfitting:** The 5-Fold Out-of-Fold (OOF) training strategy ensures that the meta-learner is trained on predictions from data the base models have never seen. This "blind" training prevents the meta-learner from overfitting to noise in the training set.
- **Optimal Weighting:** Simple averaging assumes all models are equally accurate in all scenarios. By using a Bayesian Ridge meta-learner, the system dynamically calculates probabilistic weights for each model, effectively suppressing redundant predictions and maximizing final accuracy.

What other alternatives were considered?

During the development of this pipeline, we experimented with several alternative techniques to refine the performance:

- **Single Model Reliance:** Initially, we evaluated standalone CatBoost and XGBoost models. While strong, they were prone to performance plateaus. Stacking these models proved to be the most effective way to cross the accuracy threshold.
- **Hyperparameter Grid Search:** We tested exhaustive grid search for hyperparameter tuning. However, we found that with a robust OOF stacking architecture, structural diversity (model ensembling) provided significantly better gains than marginal hyperparameter optimization.
- **Standard K-Fold vs. Target Encoding:** Early iterations used simple label encoding for categorical variables. We transitioned to Target Encoding applied *within* the cross-validation folds to mathematically handle high-cardinality geohash data without leaking target information.
- **Different Meta-Learners:** We experimented with Linear Regression and Random Forest meta-learners. Bayesian Ridge was ultimately selected for its ability to handle

multicollinearity and provide stable, probabilistic weightings that are less sensitive to outliers.

Methodology

- **Spatial-Temporal Synthesis:** Geographic coordinates are decomposed into Principal Components (PCA) to extract latent traffic patterns. Time is encoded using trigonometric (sine/cosine) functions, capturing periodic flow dynamics as a continuous cycle.
- **Leak-Free Pipeline:** Target encoding is applied strictly within each validation fold, ensuring that the base models do not have access to the target values of the validation sets.
- **Mathematical Constraints:** Target transformation (Log1p) is applied to handle the skewness of traffic demand, and final predictions are clipped at zero to enforce physical reality.

Conclusion

This pipeline moves beyond simple heuristic modeling. By combining sophisticated spatial-temporal feature engineering with a leak-free OOF stacking architecture, the model is optimized to minimize both bias and variance, ensuring that the resulting demand forecasts are reliable, generalizable, and mathematically resilient.